

# Chien-Hsien Chu

FULL-STACK • DEVOPS • PLATFORM ENGINEER • 5+ YEARS WORK EXPERIENCE

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## Education

### National Yang Ming Chiao Tung University (NYCU)

Hsinchu, Taiwan

M.S. IN COMPUTER SCIENCE

Sep. 2023 – Sep. 2026

- Overall GPA: 4.2/4.3 · Advisor: Prof. Chien-Chao Tseng
- Thesis: *Observability-Driven Root-Cause Analysis and Recovery Framework for Cloud-Native 5G Core Network Functions.*

## Academic & Technical Experience

### Observability-Driven Root-Cause Analysis and Recovery Framework for Cloud-Native 5G Core Network Functions

M.S. Thesis

GRADUATE RESEARCH ASSISTANT, WIRELESS & INTERNET LAB (WINLAB), NYCU

Feb. 2026 – Sep. 2026

- Built an End-to-End Observability and Auto-Remediation Platform for multi-cluster **Kubernetes**: Non-Intrusive Telemetry Collection and Correlation, Fault Detection, Root-Cause Analysis, and Guarded Remediation.
- Telemetry Collection and Generation: **eBPF** auto-instrumentation (**Grafana Beyla**) yielding traces and RED Metrics with no code changes, shipped over **OTLP** by **OpenTelemetry Collectors**.
- Telemetry Storage and Fault Detection: **Prometheus**, **Loki**, **Tempo** on **MinIO**; **Grafana**, **Alertmanager**.
- LLM-Based RCA Agent: a bounded investigation loop over live telemetry, guided by injected environment knowledge.
- RAG-Based Recovery Agent: Knowledge Base retrieval, then **Auto** / **Approval** / **Escalation** remediation over **MCP**.
- Transparent State-Recovery Proxy: record-and-replay recovery of a stateful service's in-memory state after a restart.

### 🔗 Infrastructure Automation & Management for Laboratory Environment

🔗 [source code](#)

SYSTEM ADMINISTRATOR, WIRELESS & INTERNET LAB (WINLAB), NYCU

Jul. 2024 – Sep. 2026

- Independently migrated a Kubernetes platform from **Kubespray** + **Terraform** to an immutable, GitOps Architecture using **Talos Linux** and **Flux CD**, including cluster bootstrap, node lifecycle automation, and PV-level data migration.
- Executed multi-application, multi-version upgrades involving rollbackable database upgrades and data migrations.
- Integrated **Velero**, **MinIO**, **Grafana Stack**, and **Keycloak** for backup, object storage, observability, and access control.
- Managed **Harvester** clusters and migrated to **Proxmox VE** for improved stability and flexibility.

### Kubernetes Networking and Security Enhancement for Department Infrastructure

SYSTEM TEACHING ASSISTANT, CSIT, NYCU

Jul. 2024 – Sep. 2026

- Migrated cluster networking from **Flannel** to **Calico**, enforcing fine-grained **NetworkPolicy** traffic isolation.
- Deployed **Istio Egress Gateway** to centralize outbound routing, policy enforcement, and auditability.
- Evaluated **External Secrets Operator (ESO)** and **OpenBao (Vault fork)** for cluster-wide secrets management.

### Multi-Cluster Management System

GRADUATE STUDENT ASSISTANT, WIRELESS & INTERNET LAB (WINLAB), NYCU

Sep. 2023 – Jan. 2024

- **Multi-Cluster Management System** integrating **Karmada**, **Cilium**, and **Thanos** for orchestration and observability.

### Kubernetes Infrastructure and DevOps Workflow Development

TECHNICAL ASSISTANT, INFORMATION TECHNOLOGY SERVICE CENTER, NYCU

Sep. 2023 – Aug. 2024

- Built scalable **Kubernetes** infrastructure for institutional systems.
- Enhanced CI/CD automation using **GitLab CI** to improve delivery reliability.

### System Architecture Planning and Azure Infrastructure Consultation

TECHNICAL CONSULTANT, KOI STUDIO LTD.

Dec. 2023 – Nov. 2024

- Advised on system architecture and **Azure** infrastructure design for a software product company.
- Resolved deployment and integration issues between cloud services and external vendor APIs.

## Work Experience

ISCOM Online International Information Inc.

R&D ENGINEER, R&D DEPARTMENT

Taichung, Taiwan  
Jul. 2022 – Aug. 2023

- Built and operated production **Kubernetes** clusters across on-premises and cloud environments (**AKS**).
- Designed and implemented internal **GitLab CI/CD Pipelines** to automate build, test, and deployment workflows.
- Established a Unified Frontend Framework with **Next.js**, enhancing development efficiency and code reuse.

FULL-STACK ENGINEER / TECHNICAL PROJECT MANAGER, BUSINESS UNIT

Jul. 2018 – Jun. 2022

- Led and delivered over **45 web and enterprise projects**, including **E-Commerce**, **POS**, and **GIS-Based Systems**.
- Developed full-stack solutions using **ASP.NET Core**, **Laravel**, **Next.js**, and **Vue.js** with **MSSQL** and **MySQL** databases.
- Integrated multiple third-party and enterprise services such as **Payment Gateways**, **LINE OA**, and **POS Systems**.
- Refactored legacy **.NET Framework** backends into cross-platform, containerizable **.NET Core**.
- Supported clients on **Azure Cloud** deployment of web apps, VMs, and databases.

Internship

SOFTWARE ENGINEER INTERN

Taichung, Taiwan  
Jul. 2015 – Jun. 2018

- Three successive internships at IT companies: **ISCOM Inc.**, **BitCloud Co.**, and **RCE Power Inc.**
- Web application development, front-end performance work, and internal data-collection automation.

## Technical Skills

|                                 |                                                                                                              |
|---------------------------------|--------------------------------------------------------------------------------------------------------------|
| Cloud & Container Orchestration | Kubernetes (AKS, Talos Linux, Kubespray), Docker, Helm, Flux CD, Harbor, Karmada                             |
| Observability                   | Prometheus, Grafana, Loki, Tempo, Thanos, OpenTelemetry (OTLP), Alertmanager                                 |
| AI & LLM Systems                | LLM Agents, RAG, MCP, Prompt Engineering                                                                     |
| Networking & Systems            | Linux System Administration, IPv4 Networking, eBPF, MetallB, Traefik, Istio, Calico, Cilium                  |
| Infrastructure & Automation     | Proxmox VE, Harvester, Terraform (OpenTofu), Ansible, Velero, GitLab CI/CD, GitOps                           |
| Security & Identity             | Keycloak, HashiCorp Vault (OpenBao), NetworkPolicy, Cert-Manager, OAuth2 / OIDC                              |
| Programming & Scripting         | Bash, Python, Go, C++, C# (.NET MVC/Core), PHP (Laravel), JavaScript (Vue.js / Next.js), SQL (MSSQL / MySQL) |

## Publications

### An O-RAN Compliant Intent-Based MANO Framework for End-to-End Network Slicing

YI-HONG LIN\*, CHIEN-HSIEN CHU, WEN-JU CHIANG, AND CHIEN-CHAO TSENG

- BIOTC 2025 (oral), IIHMSP 2025 (poster).

## Invited Talks

|           |                                                                                         |      |
|-----------|-----------------------------------------------------------------------------------------|------|
| Dec. 2025 | <b>Guest Speaker</b> , Advanced Kubernetes Networking and Troubleshooting               | NYCU |
| Sep. 2024 | <b>Guest Speaker</b> , Introduction to Kubernetes and Its Networking Fundamentals       | NYCU |
| Mar. 2022 | <b>Invited Lecturer</b> , Exploring Trends and Career Paths in the Information Industry | CSMU |

## Awards & Honors

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|------------------------------------------------------------------------------------|-------------|
| <b>Academic Excellence Award</b> , National Yang Ming Chiao Tung University (NYCU) | Spring 2024 |
| <b>Academic Excellence Award</b> , National Yang Ming Chiao Tung University (NYCU) | Fall 2023   |